

Thai Farmer LINE LLM Advisory Agent

Technical Data Architecture & System Reference — Retrieval-Grounded, Guardrailed Conversational AI over LINE for Thai smallholder farmers

v0.1 • 2026-07-16 FastAPI · PostgreSQL · pgvector Claude / OpenAI-compatible LLM RAG grounding + citations PDPA (TH) compliant Deployed on Render (SG)

< 3 s

p95 reply latency target

100%

domain answers carry citations

0

ungrounded high-risk answers

≥ 0.18

min retrieval score gate

1 SYSTEM CONTEXT & DATA FLOW

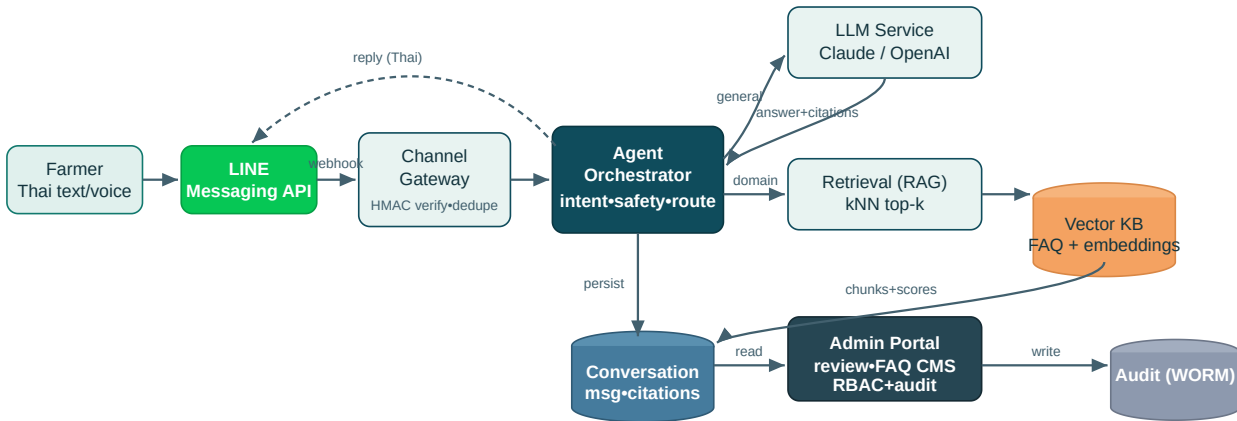


Fig 1 — C4 container view. Stateless gateway/orchestrator; strong-consistency stores for conversation & knowledge; append-only audit.

2 INBOUND REQUEST LIFECYCLE

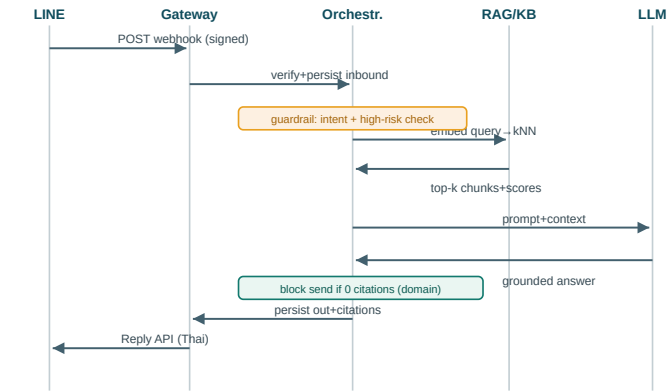


Fig 2 — Idempotent on `line_message_id`; durable write before Reply API call.

3 GUARDRAIL DECISION LOGIC

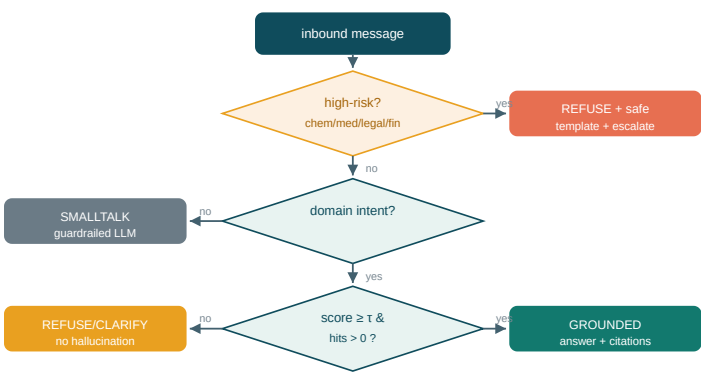


Fig 3 — Policy enforced in orchestrator, independent of model output (τ = RETRIEVAL_MIN_SCORE).

4 MESSAGE ROUTING SEMANTICS

Route	Trigger	Grounding	Citations	LLM temp	Persisted signals
smalltalk	non-domain intent	none	0	0.2	intent, model, latency
faq_grounded	domain + score $\geq\tau$	top-k FAQ chunks	≥1 (enforced)	0.2	confidence, citations[], tokens
refused	domain + weak/no match	n/a	0	—	confidence=0, reason
refused (risk)	high-risk keyword hit	n/a	0	—	intent=high_risk
escalated	flagged / human handoff	optional	optional	—	escalation ref

5 LOGICAL DATA MODEL (ERD)

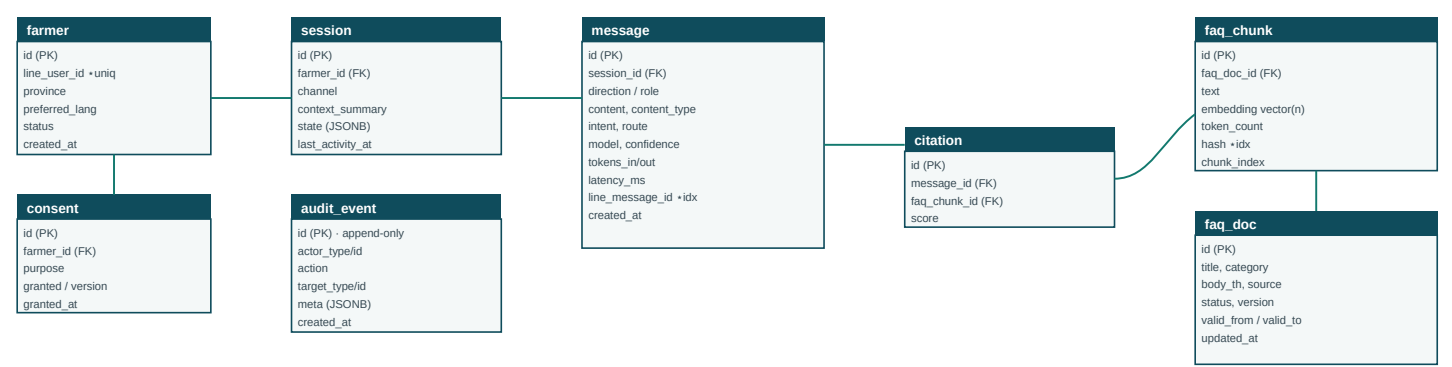
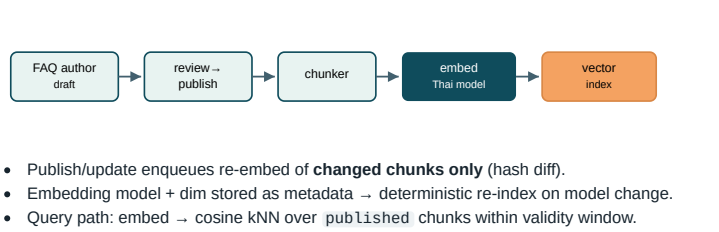


Fig 4 — UUID PKs (string-portable SQLite → Postgres). `faq_chunk.embedding` = JSON array on SQLite / native `pgvector` in prod. Erasure cascades farmer → session → message → citation + media/embeddings; audit retained lawfully.

6 KEY FIELD SEMANTICS

Field	Type	Notes
line_user_id	text	pseudonymous; no phone/national ID
message.route	enum	smalltalk-faq_grounded-refused-escalated
message.confidence	float	top retrieval cosine score
faq_doc.status	enum	only published retrievable
faq_doc.valid_*	date	seasonal/regulatory validity window
faq_chunk.hash	sha256	skip re-embed if unchanged
consent.purpose	enum	service-analytics-contact (PDPA)

7 RAG INDEXING PIPELINE



8 API CONTRACTS (SELECTED)

POST /webhook/line — LINE inbound

```
POST /webhook/line
X-Line-Signature: <base64 hmac-sha256>
{
  "events": [{
    "type": "message", "replyToken": "...",
    "source": { "userId": "U..." },
    "message": { "id": "...", "type": "text", "text": "..." }
  }]
}
→ 200 {"status": "ok", "handled": 1}
→ 403 invalid signature
```

POST /webhook/simulate — grounded reply

```
{
  "reply": "จากข้อมูลในคำถามที่: ...",
  "intent": "domain", "route": "faq_grounded",
  "confidence": 0.36, "model": "anthropic:claude-...",
  "latency_ms": 842,
  "citations": [
    { "faq_chunk_id": "...", "score": 0.36 }
  ]
}
```

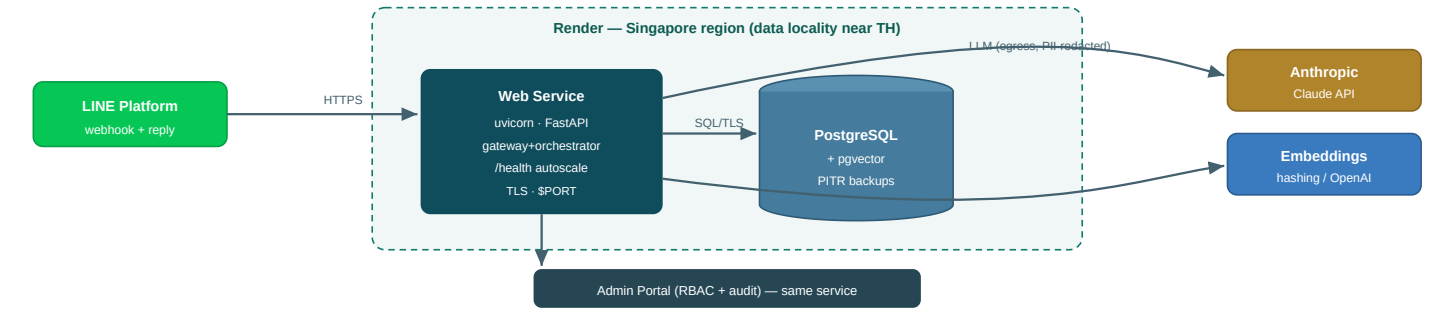


Fig 5 — Blueprint-provisioned (render.yaml). Secrets (LINE, Anthropic) injected as env; ADMIN_TOKEN auto-generated; DATABASE_URL bound from managed DB.

10 TECHNOLOGY STACK

Layer	Choice	Rationale
Runtime	Python 3.13, FastAPI, uvicorn	async I/O, typed, fast
ORM	SQLAlchemy 2.0	SQLite ↔ Postgres portability
DB	PostgreSQL + pgvector	ACID + native kNN
Cache	Redis	session ctx, dedupe, rate-limit
LLM	Claude (Messages API)	Thai quality; provider-swappable
Embeddings	hashing / OpenAI	offline dev → prod model
Channel	LINE Messaging API	primary farmer channel
Host	Render (SG) IaC blueprint	managed TLS, DB, autodeploy

11 SLOS & OBSERVABILITY

Signal	Target / captured
Reply latency p95	< 3 s end-to-end
Grounding rate	100% of faq_grounding have ≥1 citation
Availability	99.5% monthly (stateless web tier)
Per-message metrics	route, intent, confidence, tokens_in/out, latency_ms, model
Retrieval quality	offline eval set; score distribution monitoring
Cost	token accounting per message → per-conversation cost

Anti-misinformation invariants — (1) high-risk ⇒ never generated; (2) domain answer ⇒ grounded + cited or refused; (3) only published, in-validity FAQ retrievable; (4) source attribution surfaced to farmer.

12 SECURITY & PDPA (THAILAND)

- **Consent:** purpose-based, versioned, captured at LINE onboarding.
- **Minimization:** pseudonymous line_user_id; no phone/national ID.
- **Webhook integrity:** HMAC-SHA256 signature verify + idempotency dedupe.
- **Erasure/portability:** cascade delete + time-limited export; audit retained lawfully.
- **RBAC:** viewer/editor/dpo; PII view logged to append-only audit.
- **Residency:** in-region stores + LLM egress PII-redacted; DPA with processors.
- **Encryption:** TLS in transit; at-rest DB encryption; secrets in env vault.

13 RETENTION & RISK REGISTER

Data	Hot	Then
message content	90d	archive 1y → delete
media	30d	delete w/ message
session (Redis)	30m idle	auto-expire
audit	1–2y	WORM

Risk	Mitigation
hallucinated advice	grounded-only + citations
stale seasonal FAQ	validity dates + review
PII → external LLM	redaction + DPA + region
webhook replay	signature + idempotency